

Week 5: Sentiment Analysis Dashboard - MindPulse AI

“Tracking Emotions. Empowering Well-being.”

General Overview

MindPulse AI is an AI-powered sentiment and emotion analytics dashboard developed to monitor and interpret emotional well-being trends across social media platforms, surveys, and community spaces.

Tool choice

The no-code framework Base44 was used for building the application which allowed for fast development time without compromising scalability or performance. The application was then also deployed using Base44's built in deployment.

Usage

Users may upload individual text entries or images for personal emotion assessment. The system's AI Wellness Assistant provides supportive resources or motivational prompts when elevated stress levels are detected, while the Mood Prediction Engine forecasts future emotional trends based on historical data.

The platform is particularly useful for universities tracking student sentiment during examinations, organizations monitoring workplace morale, and public health researchers studying collective mental well-being.

Key Learnings

1. **Data Ethics in AI Development:** Integrating anonymization, consent mechanisms, and compliance protocols (GDPR/POPIA) was essential to maintain trust and safeguard users' psychological data.
2. **Multimodal Emotion Detection Complexity:** Combining textual and visual emotion analysis required robust preprocessing pipelines and attention to cultural variations in emotional expression.
3. **User Experience in Mental Health Interfaces:** Designing a calm, minimal, and non-triggering interface improved user engagement and ensured emotional safety for those interacting with well-being data.
4. **Interdisciplinary Insight:** The project highlighted the intersection between AI engineering, psychology, and user interface design, reinforcing the value of cross-domain collaboration.